A BASIS FOR A COMPLEMENT OF ONE FILTRATION LAYER OF THE ASKEY-WILSON ALGEBRA IN THE NEXT, BUILT FROM OMEGA POWERS

PAUL TERWILLIGER

Editorial note. All delivered mathematical text is the author's own, copied from the pinned publisher source sources/sigma-2011-069-askey-wilson/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the retained preamble is the source's own layout adapter plus the author's macro block at lines 10-18 of sources/sigma-2011-069-askey-wilson/pinned-original-source.tex (everything between the publisher class metadata and \begin{document}, comments excluded). Nothing was added, no macro was invented, and the publisher document class is deliberately not reproduced; the excerpt serves only as preamble, never as a span coordinate.

Editorial note. External references are navigation pointers only; every deliberately kept step of the author's argument stays inside the counted spans. The displayed numbers of those pointers are the author's own numbering under the author's \newtheorem declarations (the source has no subsections, so numbering is section.index), computed mechanically by the lane's scanner.

Editorial note. Three references to other labelled items occur inside the span, Corollary lem:com and Lemmas lem:omegapower and lem:almostcom; each is recorded once as a navigation stub with the matching kind and the author's own number, and every step that consumes them stays inside the counted span.

Editorial note. The counted spans contain no bibliography citation, so no bibliography excerpt is attached to this card.

Editorial note. Licence binding: the card's source_version now carries the value recorded in this source's own licence evidence (sources/sigma-2011-069-askey-wilson/license-observation.json), namely 'SIGMA7(2011),069; publisher final form2011-07-09'. The adopt step of the publication line compares the two and refused the earlier revision; the card is aligned to the evidence rather than the evidence to the card.

Allowed standard prerequisites (editorial).

- A monomial basis for a complement of Delta_{n-1} in Delta_n. Same source, Corollary labelled lem:com; invoked inside the counted proof at lines 1676-1683. Supplies the six-tuple basis indexed by I_n against which the Omega-power basis is compared.
- Expressing a power of Omega in terms of monomials modulo the previous layer. Same source, Lemma labelled lem:omegapower; invoked inside the counted proof at lines 1696-1702. Replaces Omega^ell by q^{ell^2} A^ell B^ell C^ell modulo Delta_{n-1}.
- Moving the letters A, B, C past each other modulo the previous layer. Same source, Lemma labelled lem:almostcom; invoked inside the counted proof at lines 1703-1708. Converts A^ell B^ell C^ell into A^i B^j C^k form with the numerical factor q^{2 j ell}.

Date: doi:10.3842/SIGMA.2011.069; CC BY4.0.

Source reference. Original lemma 1.25: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 1703–1703. Navigation pointer for \ref{lem:almostcom} used inside the counted span (line 1703); the argument that consumes it is retained in the span.

Source reference. Original corollary 1.27: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 1676–1676. Navigation pointer for \ref{lem:com} used inside the counted span (line 1676); the argument that consumes it is retained in the span.

Source reference. Original lemma 1.32: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 1697–1697. Navigation pointer for \ref{lem:omegapower} used inside the counted span (line 1697); the argument that consumes it is retained in the span.

Lemma 0.1. *For all integers $n \geq 1$ the following is a basis for a complement of Δ_{n-1} in Δ_n:*

$$A^i B^j C^k \Omega^\ell \alpha^r \beta^s \gamma^t, \qquad i,j,k,\ell,r,s,t \geq 0, \qquad ijk = 0, \qquad i+j+k+3\ell+r+s+t = n.$$

Proof. Let $\mathbb{I}_n$ denote the set consisting of the 6-tuples (i,j,k,r,s,t) of nonnegative integers whose sum is n. By Corollary 1.27 the following is a basis for a complement of Δ_{n-1} in Δ_n:

$$A^i B^j C^k \alpha^r \beta^s \gamma^t, \qquad (i,j,k,r,s,t) \in \mathbb{I}_n.$$

Let $\mathbb{J}_n$ denote the set consisting of the 7-tuples (i,j,k,ℓ,r,s,t) of nonnegative integers such that $ijk = 0$ and $i+j+k+3\ell+r+s+t = n$. Observe that the map

$$\mathbb{J}_n \to \mathbb{I}_n, \qquad (i,j,k,\ell,r,s,t) \mapsto (i+\ell, j+\ell, k+\ell, r, s, t)$$

is a bijection. Suppose we are given $(i,j,k,\ell,r,s,t) \in \mathbb{J}_n$. By Lemma 1.32, Δ_{n-1} contains

$$A^i B^j C^k \Omega^\ell \alpha^r \beta^s \gamma^t - q^{\ell^2} A^i B^j C^k A^\ell B^\ell C^\ell \alpha^r \beta^s \gamma^t.$$

By Lemma 1.25, Δ_{n-1} contains

$$A^i B^j C^k A^\ell B^\ell C^\ell \alpha^r \beta^s \gamma^t - q^{2j\ell} A^{i+\ell} B^{j+\ell} C^{k+\ell} \alpha^r \beta^s \gamma^t.$$

Therefore Δ_{n-1} contains

$$A^i B^j C^k \Omega^\ell \alpha^r \beta^s \gamma^t - q^{\ell(2j+\ell)} A^{i+\ell} B^{j+\ell} C^{k+\ell} \alpha^r \beta^s \gamma^t.$$

By these comments the following is a basis for a complement of Δ_{n-1} in Δ_n:

$$A^i B^j C^k \Omega^\ell \alpha^r \beta^s \gamma^t, \qquad (i,j,k,\ell,r,s,t) \in \mathbb{J}_n.$$

The result follows. □

References

[1] Alperin R.C., Notes: $\mathrm{PSL}_2(\mathbb{Z}) = \mathbb{Z}_2 \star \mathbb{Z}_3$, *Amer. Math. Monthly* **100** (1993), 385–386.